

Quantitative Aptitude

Topics Covered:

1. Percentage
2. Profit & Loss
3. Ratio & Proportion
4. Average
5. Time & Work
6. Time, Speed & Distance
7. Simple & Compound Interest
8. Probability
9. Permutation & Combination
10. Number System
11. Geometry
12. Data Interpretation
13. Additional

1. Percentage

- $\text{Percentage} = (\text{Part} / \text{Whole}) \times 100$
- $\text{Value} = (\text{Percentage} \times \text{Total}) / 100$
- $\text{Percentage Increase} = ((\text{New} - \text{Old}) / \text{Old}) \times 100$
- $\text{Percentage Decrease} = ((\text{Old} - \text{New}) / \text{Old}) \times 100$

Shortcut

- Increase by $x\% \rightarrow$ Multiply by $(100 + x)/100$
- Decrease by $x\% \rightarrow$ Multiply by $(100 - x)/100$

2. Profit and Loss

- $\text{Profit} = \text{SP} - \text{CP}$
- $\text{Loss} = \text{CP} - \text{SP}$
- $\text{Profit \%} = (\text{Profit} / \text{CP}) \times 100$
- $\text{Loss \%} = (\text{Loss} / \text{CP}) \times 100$
- $\text{SP} = \text{CP} \times (100 + \text{Profit\%})/100$
- $\text{SP} = \text{CP} \times (100 - \text{Loss\%})/100$
- $\text{CP} = \text{SP} \times 100/(100 + \text{Profit\%})$
- $\text{CP} = \text{SP} \times 100/(100 - \text{Loss\%})$

3. Simple Interest (SI)

- $\text{SI} = (\text{P} \times \text{R} \times \text{T})/100$
- $\text{Amount} = \text{P} + \text{SI}$

Where:

- P = Principal
- R = Rate (%)
- T = Time (years)

4. Compound Interest (CI)

- $\text{Amount} = \text{P}(1 + \text{R}/100)^T$
- $\text{CI} = \text{Amount} - \text{Principal}$

1 Simple Interest  $SI = \frac{PRT}{100}$	2 Amount in Simple Interest  $A = P + SI$	3 Rate from SI  $R = \frac{SI \times 100}{P \times T}$	4 Time from SI  $T = \frac{SI \times 100}{P \times R}$
5 Principal from SI  $P = \frac{SI \times 100}{R \times T}$	6 Compound Amount (annual)  $A = P \left(1 + \frac{r}{100}\right)^n$	7 Compound Interest  $CI = A - P$	8 Compound Amount (half-yearly)  $A = P \left(1 + \frac{r}{200}\right)^{2n}$
9 Compound Amount (quarterly)  $A = P \left(1 + \frac{r}{400}\right)^{4n}$	10 Depreciation Formula  $A = P \left(1 - \frac{r}{100}\right)^n$	11 Population Growth / Repeated Growth  $A = P \left(1 + \frac{r}{100}\right)^n$	12 Present Value  $P = \frac{A}{\left(1 + \frac{r}{100}\right)^n}$
13 Difference between CI and SI for 2 years  $CI - SI = P \left(\frac{r}{100}\right)^2$	14 Difference between CI and SI for 3 years  $CI - SI = P \left(\frac{r}{100}\right)^2 \left(3 + \frac{r}{100}\right)$	15 If amount doubles in n years  $2 = \left(1 + \frac{r}{100}\right)^n$	16 If amount triples in n years  $3 = \left(1 + \frac{r}{100}\right)^n$
17 Effective annual rate with m compounds  $\text{Effective rate} = \left(1 + \frac{r}{100m}\right)^m - 1$	18 Successive increase x% and y%  $\text{Net \%} = x + y + \frac{xy}{100}$	19 Successive decrease x% and y%  $\text{Net decrease \%} = x + y - \frac{xy}{100}$	20 Rate from compound amount  $r = \left\{ \left[\left(\frac{A}{P} \right)^{1/n} - 1 \right] \right\} \times 100$

5. Ratio and Proportion

If $A:B = m:n$

Then

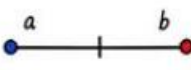
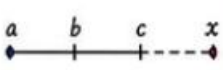
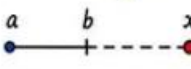
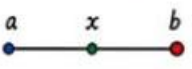
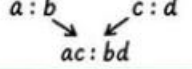
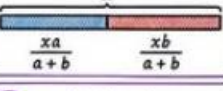
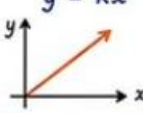
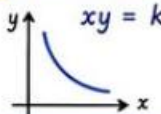
$$A = \frac{m}{m+n} \times \text{Total}$$

$$B = \frac{n}{m+n} \times \text{Total}$$

Proportion:

$A:B = C:D$

$$\Rightarrow A \times D = B \times C$$

1 Ratio form $a : b = \frac{a}{b}$ 	2 Equivalent ratio $a : b = ma : mb$ $a : b \rightarrow ma : mb$	3 Proportion rule if $a : b = c : d$, then $ad = bc$ 	4 Fourth proportional $x = \frac{bc}{a}$ 
5 Third proportional $x = \frac{b^2}{a}$ 	6 Mean proportional $x = \sqrt{ab}$ 	7 If $a : b = m : n$ $\frac{a}{a+b} = \frac{m}{m+n}$ 	8 If $a : b = m : n$ $\frac{b}{a+b} = \frac{n}{m+n}$ 
9 If $a : b = m : n$ $\frac{a-b}{a+b} = \frac{m-n}{m+n}$ 	10 Compound ratio $(a : b)$ and $(c : d)$ gives $ac : bd$ 	11 Duplicate ratio $a^2 : b^2$ $a : b \rightarrow a^2 : b^2$	12 Triplicate ratio $a^3 : b^3$ $a : b \rightarrow a^3 : b^3$
13 Subduplicate ratio $\sqrt{a} : \sqrt{b}$ $a : b \rightarrow \sqrt{a} : \sqrt{b}$	14 Continued proportion if $a : b = b : c$, then $b^2 = ac$ 	15 Divide x in ratio a : b parts = $\frac{xa}{a+b}, \frac{xb}{a+b}$ 	16 Direct proportion $y = kx$ 
17 Inverse proportion 	18 Componendo if $\frac{a}{b} = \frac{c}{d}$, then $\frac{a+b}{b} = \frac{c+d}{d}$	19 Dividendo $\frac{a-b}{b} = \frac{c-d}{d}$	20 Componendo and dividendo $\frac{a+b}{a-b} = \frac{c+d}{c-d}$ 

6. Average

- Average = Sum of observations / Number of observations
- Sum = Average $\times$ Number

7. Time and Work

If A finishes work in x days

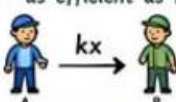
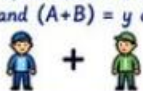
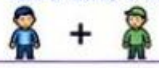
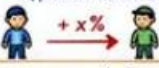
One day's work = $1/x$

Together

$$1/T = 1/A + 1/B$$

For three people

$$1/T = 1/A + 1/B + 1/C$$

1 One-day work  If A finishes in x days, one-day work = $\frac{1}{x}$	2 Total days  Days = $\frac{1}{\text{(one-day work)}}$	3 A and B one-day work  $\frac{1}{x} + \frac{1}{y}$	4 A and B together time  $\frac{xy}{x+y}$
5 Pipe fill and drain net work  $\frac{1}{x} - \frac{1}{y}$	6 If A is k times as efficient as B  time ratio A:B = 1:k	7 Efficiency and time inverse relation  $E_A : E_B = y : x$	8 If efficiency ratio is m:n  time ratio = n:m
9 For same work  Men $\times$ Days = constant	10 For same work  Men $\times$ Hours $\times$ Days = constant	11 If A alone = x days and (A+B) = y days  B alone = $\frac{xy}{x-y}$	12 A + B + C one-day work  $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$
13 A + B + C together time  $\frac{xyz}{xy + yz + zx}$	14 Alternate day work of A and B  two-day work = $\frac{1}{x} + \frac{1}{y}$	15 If m men finish in d days  one man-day work = $\frac{1}{md}$	16 Wages rule  Wages $\propto$ work done
17 If efficiency increases by x%  time factor = $\frac{100}{100+x}$	18 If time decreases by x%  efficiency factor = $\frac{100}{100-x}$	19 If A works a days and B works b days  work done = $\frac{a}{x} + \frac{b}{y}$	20 If A is x% more efficient than B  time ratio A:B = 100 : (100 + x)

8. Pipes and Cisterns

Same as Time and Work.

Filling pipe = Positive

Leak pipe = Negative

Total Rate = Sum of all rates

9. Time, Speed and Distance

Distance = Speed $\times$ Time

Speed = Distance / Time

Time = Distance / Speed

Conversions

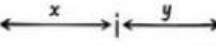
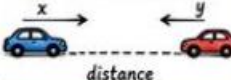
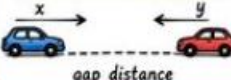
- 1 km/hr = 5/18 m/s

- 1 m/s = 18/5 km/hr

Average Speed

Equal distances

Average Speed = $2xy/(x+y)$

1 Speed $\text{Speed} = \frac{\text{Distance}}{\text{Time}}$ 	2 Distance $\text{Distance} = \text{Speed} \times \text{Time}$ 	3 Time $\text{Time} = \frac{\text{Distance}}{\text{Speed}}$ 	4 km/h to m/s multiply by $\frac{5}{18}$  km/h → m/s
5 m/s to km/h multiply by $\frac{18}{5}$  m/s → km/h	6 Average speed for equal distances $\frac{2xy}{x+y}$ 	7 Relative speed opposite direction $x + y$ 	8 Relative speed same direction $ x - y $ 
9 Train crossing a pole $\text{time} = \frac{\text{train length}}{\text{speed}}$ 	10 Train crossing a platform $\text{time} = \frac{\text{train length} + \text{platform length}}{\text{speed}}$ 	11 Boat downstream speed $b + s$ 	12 Boat upstream speed $b - s$ 
13 Speed in still water $b = \frac{\text{downstream} + \text{upstream}}{2}$ 	14 Stream speed $s = \frac{\text{downstream} - \text{upstream}}{2}$ 	15 Fixed distance rule $\text{speed} \times \text{time} = \text{constant}$ 	16 If speed increases by x% $\text{new time factor} = \frac{100}{100 + x}$ 
17 If speed decreases by x% $\text{new time factor} = \frac{100}{100 - x}$ 	18 Average speed for equal times $\frac{x + y}{2}$ 	19 Meeting time $\text{time} = \frac{\text{distance}}{x + y}$ 	20 Catching time $\text{time} = \frac{\text{gap distance}}{x - y}$ 

10. Trains

Train crossing pole

Time = Train Length / Speed

Two trains opposite directions

Relative Speed = Speed₁ + Speed₂

Same direction

Relative Speed = Difference of speeds

11. Boats and Streams

Downstream

$$= \text{Boat Speed} + \text{Stream Speed}$$

Upstream

$$= \text{Boat Speed} - \text{Stream Speed}$$

Boat Speed

$$= (\text{Downstream} + \text{Upstream})/2$$

Stream Speed

$$= (\text{Downstream} - \text{Upstream})/2$$

12. Mixture and Alligation

Weighted Average

$$\text{Mean} = (a \times w_1 + b \times w_2)/(w_1 + w_2)$$

13. Partnership

Profit Share

Investment $\times$ Time

Example

A invests ₹1000 for 12 months

B invests ₹2000 for 6 months

Ratio

$$1000 \times 12 : 2000 \times 6$$

$$= 1 : 1$$

14. Ages

Present Age $\pm$ Years

Future age

Present + Years

Past age

Present – Years

15. Permutation

Permutation = Arrangement matters (Order matters).

$${}^nP_r = \frac{n!}{(n-r)!}$$

Question:

From 5 students (A, B, C, D, E), **how many ways can you choose and arrange 3 students** to stand in a line?

Here, changing the order creates a different arrangement.

For example:

- ABC
- ACB
- BAC

These are all **different**.

Here,

- $n = 5$
- $r = 3$

$${}^5P_3 = \frac{5!}{(5-3)!} = \frac{5!}{2!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = 60$$

16. Combination

Combination = Selection only (Order does not matter).

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

Question:

From the same 5 students, **how many ways can you choose 3 students** to form a team?

Now the order does **not** matter.

For example:

- ABC
- ACB
- BAC

These all represent the **same team**.

Formula:

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

Here,

$${}^5C_3 = \frac{5!}{3!2!} = \frac{120}{6 \times 2} = \frac{120}{12} = 10$$

17. Probability

$$P(E) = \frac{\text{Favourable Outcomes}}{\text{Total Outcomes}}$$

Complement

P(Not E)

$$= 1 - P(E)$$

18. Number System

Even numbers

$$2n$$

Odd numbers

$$2n+1$$

Sum of first n natural numbers

$$\frac{n(n+1)}{2}$$

Sum of squares

$$\frac{n(n+1)(2n+1)}{6}$$

Sum of cubes

$$\left(\frac{n(n+1)}{2}\right)^2$$

19. HCF and LCM

For two numbers

HCF $\times$ LCM = Product of Numbers

20. Quadratic Equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

22. Algebraic Identities

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$a^2 - b^2 = (a + b)(a - b)$$

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

23. Geometry

Rectangle

- Area = Length $\times$ Breadth
- Perimeter = 2(L+B)

Square

- Area = Side²
- Perimeter = 4 $\times$ Side
- Diagonal = Side $\sqrt{2}$

Triangle

- Area = $\frac{1}{2} \times$ Base $\times$ Height

Circle

- Circumference = $2\pi r$
- Area = πr^2

Cylinder

- Volume = $\pi r^2 h$
- CSA = $2\pi r h$
- TSA = $2\pi r(h+r)$

Cone

- Volume = $(\frac{1}{3})\pi r^2 h$

Sphere

- Surface Area = $4\pi r^2$
- Volume = $(\frac{4}{3})\pi r^3$

24. Data Interpretation

Mean

$$= \Sigma X / n$$

Median

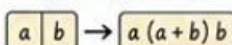
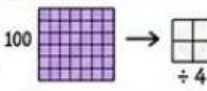
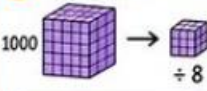
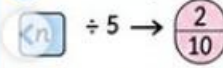
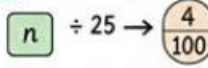
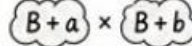
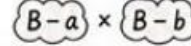
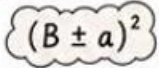
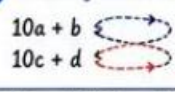
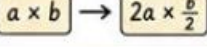
- Odd observations = Middle value
- Even observations = Average of two middle values

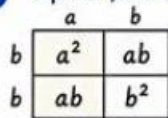
Mode

= Most frequent value

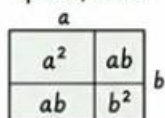
Percentage Change

$$= ((\text{New} - \text{Old}) / \text{Old}) \times 100$$

1 Complement from 10  $\text{complement} = 10 - n$	2 Complement from 100  $\text{complement} = 100 - n$	3 Complement from 1000  $\text{complement} = 1000 - n$	4 Multiply by 9  $n \times 9 = n \times 10 - n$
5 Multiply by 99  $n \times 99 = n \times 100 - n$	6 Multiply by 11 (two-digit number)  insert the sum of digits in the middle	7 Multiply by 25  $n \times 25 = \frac{n \times 100}{4}$	8 Multiply by 125  $n \times 125 = \frac{n \times 1000}{8}$
9 Divide by 5  $n \div 5 = \frac{2n}{10}$	10 Divide by 25  $n \div 25 = \frac{4n}{100}$	11 Square of numbers ending in 5 $n5^2 \rightarrow n(n+1)25$ $n5^2 = n(n+1)25$	12 Same tens, units sum 10  $(10a+b)(10a+c) = a(a+1)bc$ where $b+c=10$
13 Same units, tens sum 10  $(10a+b)(10c+b) = (ac+b)b^2$ where $a+c=10$	14 Near-base multiplication above base  $(B+a)(B+b) = B(B+a+b) + ab$	15 Near-base multiplication below base  $(B-a)(B-b) = B(B-a-b) + ab$	16 Near-base squaring  $(B \pm a)^2 = B^2 \pm 2Ba + a^2$
17 Vertically and crosswise  $(10a+b)(10c+d) = 100ac + 10(ad+bc) + bd$	18 Subtraction from powers of 10 	19 Doubling and halving trick  $(a \times b) = (2a \times \frac{b}{2})$	20 Casting out 9s  digit sum check for verification

1 Square formula

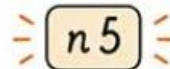
$$(a+b)^2 = a^2 + 2ab + b^2$$

2 Square formula

$$(a-b)^2 = a^2 - 2ab + b^2$$

3 Difference of squares

$$a^2 - b^2 = (a+b)(a-b)$$

4 Numbers ending in 5

$$n5^2 = n(n+1) \mid 25$$

5 Two-digit square

$$10a + b$$

$$(10a + b)^2 = 100a^2 + 20ab + b^2$$

6 Near 100 square

$$100 + a$$

$$(100 + a)^2 = 10000 + 200a + a^2$$

7 Next square from previous

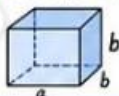
$$(n-1)^2 \rightarrow n^2$$

$$n^2 = (n-1)^2 + 2n - 1$$

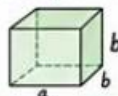
8 Difference of consecutive squares

$$n^2 - (n+1)^2$$

$$(n+1)^2 - n^2 = 2n + 1$$

9 Cube formula

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

10 Cube formula

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

11 Sum of cubes factorization

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

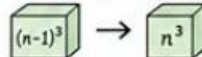
12 Difference of cubes factorization

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

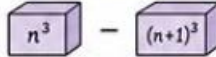
13 Two-digit cube

$$10a + b$$

$$(10a + b)^3 = 1000a^3 + 300a^2b + 30ab^2 + b^3$$

14 Next cube from previous

$$n^3 = (n-1)^3 + 3n^2 - 3n + 1$$

15 Difference of consecutive cubes

$$(n+1)^3 - n^3 = 3n^2 + 3n + 1$$

16 Square root of a perfect square

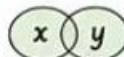
$$\sqrt{144}$$

pair prime factors and take one from each pair

17 Cube root of a perfect cube

$$\sqrt[3]{216}$$

group prime factors in triplets

18 If $x + y = s$ and $xy = p$ 

$$x^2 + y^2 = s^2 - 2p$$

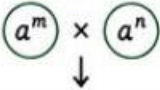
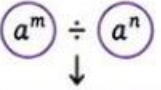
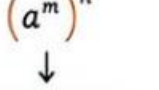
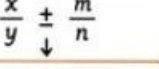
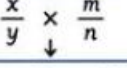
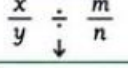
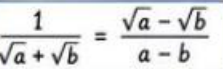
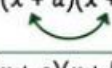
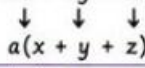
19 If $x + y = s$ and $xy = p$ 

$$x^3 + y^3 = s^3 - 3ps$$

20 Near-base square

$$B - a$$

$$(B-a)^2 = B^2 - 2Ba + a^2$$

1 BODMAS / order of operations B Brackets $()$ O Orders $\sqrt{x}$ D Division $\div$ M Multiplication $\times$ A Addition $+$ S Subtraction $-$	2 Same-base multiplication  $a^m \times a^n = a^{m+n}$	3 Same-base division  $\frac{a^m}{a^n} = a^{m-n}$	4 Power of a power  $(a^m)^n = a^{mn}$
5 Product power rule $(ab)^n$  $(ab)^n = a^n b^n$	6 Quotient power rule $\left(\frac{a}{b}\right)^n$  $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$	7 Square identity  $(a+b)^2 = a^2 + 2ab + b^2$	8 Square identity  $(a-b)^2 = a^2 - 2ab + b^2$
9 Difference of squares  $a^2 - b^2 = (a+b)(a-b)$	10 Cube identity  $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$	11 Cube identity  $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$	12 Fraction addition/subtraction $\frac{x}{y} \pm \frac{m}{n}$  $\frac{x}{y} \pm \frac{m}{n} = \frac{xn \pm ym}{yn}$
13 Fraction multiplication $\frac{x}{y} \times \frac{m}{n}$  $\frac{x}{y} \times \frac{m}{n} = \frac{xm}{yn}$	14 Fraction division $\frac{x}{y} \div \frac{m}{n}$  $\frac{x}{y} \div \frac{m}{n} = \frac{xn}{ym}$	15 Surd product $\sqrt{a} \times \sqrt{b}$  $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$	16 Surd quotient $\frac{\sqrt{a}}{\sqrt{b}}$  $\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$
17 Rationalization $\frac{1}{\sqrt{a} + \sqrt{b}} \rightarrow \frac{\sqrt{a} - \sqrt{b}}{a - b}$  $\frac{1}{\sqrt{a} + \sqrt{b}} = \frac{\sqrt{a} - \sqrt{b}}{a - b}$	18 Product expansion $(x+a)(x+b)$  $(x+a)(x+b) = x^2 + (a+b)x + ab$	19 Common factor rule $ax + ay + az$  $a(x + y + z)$ factor common terms first	20 Simplify ratios or fractions $24 : 36 \rightarrow 2 : 3$ divide by the HCF

1 Even + even = even

$$2k + 2m = 2n$$

2 Even + odd = odd

$$2k + 2m+1 = 2n+1$$

3 Odd + odd = even

$$2k+1 + 2m+1 = 2n$$

4 Odd $\times$ odd = odd

$$2k+1 \times 2m+1 = 2n+1$$

5 Even $\times$ odd = even

$$2k \times 2m+1 = 2n$$

6 Divisibility by 2
Last digit
divisible by 2

0 2 4 6 8

7 Divisibility by 3
Sum of digits
divisible by 3

$$1 + 4 + 5 = 10$$

$1+0=1$ (not div. by 3)
 $1+4+5=10$ (div. by 3)

8 Divisibility by 4
Last two digits
divisible by 4

$$ab \rightarrow \overline{ab}$$

divisible by 4

9 Divisibility by 5
Last digit
0 or 5

0 5

10 Divisibility by 6
Divisible by
2 and 3



11 Divisibility by 8
Last three digits
divisible by 8

$$abc \rightarrow \overline{abc}$$

divisible by 8

12 Divisibility by 9
Sum of digits
divisible by 9

$$2 + 7 + 1 + 8 = 18$$

18 is divisible by 9

13 Divisibility by 10
Last digit
0

— 0

14 Divisibility by 11
Difference of
alternating digit sums
is 0 or multiple of 11

$$\begin{array}{cccccc} 1 & 3 & 2 & 7 & 6 \\ + & - & + & - & + \\ |(1+2+6) - (3+7)| = |9-10| = 1 \end{array}$$

(not divisible by 11)

15 Divisibility by 12
Divisible by
3 and 4



16 Sum of first n
natural numbers

$$\frac{n(n+1)}{2}$$

17 Sum of first n
odd numbers

$$n^2$$

$$1 + 3 + 5 + \dots + (2n-1) = n^2$$

18 Sum of first n
squares

$$\frac{n(n+1)(2n+1)}{6}$$

$$1^2 \quad 2^2 \quad 3^2 \quad \dots \quad n^2$$

19 Sum of first n
cubes

$$\left[\frac{n(n+1)}{2} \right]^2$$

$$1^3 \quad 2^3 \quad 3^3 \quad \dots \quad n^3$$

20 If $N = p^a \times q^b \times r^c$

Number of factors
 $= (a+1)(b+1)(c+1)$

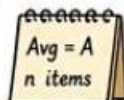
$$N \rightarrow p^a, q^b, r^c$$

1 Average



$$\text{Average} = \frac{\text{Sum}}{n}$$

2 Sum from average



$$\text{Sum} = \text{Average} \times n$$

3 Combined average



$$\frac{n_1 a_1 + n_2 a_2}{n_1 + n_2}$$

4 New average after adding x



$$\frac{nA + x}{n + 1}$$

5 New average after removing x



$$\frac{nA - x}{n - 1}$$

6 Average of first n natural numbers

1 2 3 4 ... n

$$\frac{n + 1}{2}$$

7 Average of first n even numbers

2 4 6 8 ... 2n

$$n + 1$$

8 Average of first n odd numbers

1 3 5 7 ... (2n-1)

$$n$$

9 Average of consecutive numbers

a-2 a-1 a a+1 a+2

middle number

10 Average of an AP

a, a + d, a + 2d, ..., l

$$\frac{\text{first} + \text{last}}{2}$$

11 Weighted average



$$\frac{\sum w x}{\sum w}$$

12 If each value increases or decreases by x

$x_1 \quad x_2 \quad \dots \quad x_n$
↓ ↓ ↓
 $x_1 + x \quad x_2 + x \quad \dots \quad x_n + x$

average changes by x

13 Replacing x by y

x → y

$$\text{new average} = A + \frac{y - x}{n}$$

14 Missing value

$a_1 \quad a_2 \quad \dots \quad ? \quad \dots \quad a_n$

required total - known sum

15 Omitted item

$a_1 \quad a_2 \quad \dots \quad \dots \quad a_n$

$$\text{omitted value} = nA - (n-1)A'$$

16 Added item

$a_1 \quad a_2 \quad \dots \quad a_n \quad ?$

$$\text{added value} = (n+1)A' - nA$$

17 If average increases by x in n items



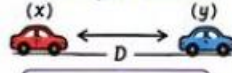
$$\text{total increase} = nx$$

18 If average decreases by x in n items



$$\text{total decrease} = nx$$

19 Average speed for equal distances

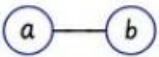
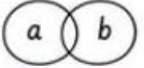


$$\frac{2xy}{x + y}$$

20 Ratio of sums



$$n_1 a_1 : n_2 a_2$$

1 For two numbers  $HCF \times LCM =$ product of the numbers	2 HCF of co-primes $(a, b) = 1$  $HCF(a, b) = 1$	3 LCM of co-primes $(a, b) = 1$  $LCM(a, b) =$ $a \times b$	4 HCF by prime factors (common, lowest powers) $a = 2^2 \times 3 \times 5$ $b = 2 \times 3^2 \times 7$ $\rightarrow$ $HCF = 2^1 \times 3^1 = 2 \times 3$
5 LCM by prime factors (all factors, highest powers) $a = 2^2 \times 3 \times 5$ $b = 2 \times 3^2 \times 7$ $\downarrow$ $LCM = 2^2 \times 3^2 \times 5 \times 7$	6 LCM(a, b) using HCF $LCM(a, b) =$ $\frac{a \times b}{HCF(a, b)}$	7 Euclid rule $HCF(a, b) =$ $HCF(b, a \bmod b)$ $(a, b) \rightarrow (b, a \bmod b) \rightarrow (r, 0)$ $\curvearrowright$ $HCF = r$	8 HCF of fractions $\frac{a}{b}, \frac{c}{d}$ $HCF = \frac{HCF(a, c)}{LCM(b, d)}$
9 LCM of fractions $\frac{a}{b}, \frac{c}{d}$ $LCM = \frac{LCM(a, c)}{HCF(b, d)}$	10 If one number divides the other $a \mid b \quad (a < b)$ $HCF(a, b) = a$ (smaller number)	11 If one number divides the other $a \mid b \quad (a < b)$ $LCM(a, b) = b$ (larger number)	12 HCF of consecutive numbers $n, n + 1$ $HCF(n, n + 1) = 1$
13 HCF of consecutive even numbers $2n, 2n + 2$ $HCF(2n, 2n + 2) = 2$	14 LCM of co-prime consecutive numbers $n, n + 1$ (co-prime) $LCM(n, n + 1) =$ $n \times (n + 1)$	15 For a^m and a^n (same base) $HCF(a^m, a^n) =$ $a^{\min(m, n)}$ $\uparrow$ $\min(m, n)$	16 For a^m and a^n (same base) $LCM(a^m, a^n) =$ $a^{\max(m, n)}$ $\uparrow$ $\max(m, n)$
17 Greatest number dividing a and b leaving same remainder r $a \equiv b \equiv r \pmod{d}$ $HCF = HCF(a - r, b - r)$	18 Least number to add to N for divisibility by d $N = dq + r$ $(0 \leq r < d)$ $Add = d - r$	19 Least number to subtract from N for divisibility by d $N = dq + r$ $(0 \leq r < d)$ $Subtract = r$	20 If $HCF = h$ and $LCM = l$ Numbers = hm and hn where m and n are co-prime and $mn = \frac{l}{h}$ 

AVERAGE FORMULAS

1 **AVERAGE** = $\frac{\text{Sum of all observations}}{\text{Number of observations}}$ 

2 Average of first n natural numbers = $\frac{(n+1)}{2}$ 

3 Average of natural numbers up to n = $\frac{(n+1)}{2}$ 

4 Average of first n even natural numbers = $\frac{n}{2}$ $2n$ 

5 Average of first n odd natural numbers = $\frac{(n+1)}{2}$ $2n-1$ 

6 Average of first n odd numbers = n n 

7 Average of first n even numbers = $n+1$ $n+1$ 

8 Average of squares of first n natural numbers = $\frac{(n+1)(2n+1)}{6}$ x^2 

9 Average of cubes of first n natural numbers = $\frac{n(n+1)^2}{4}$ x^3 

10 If the number of observations is even, then
Average = $\frac{\text{First middle term} + \text{Last middle term}}{2}$ 

11 Change in value of a quantity = $\frac{(\text{New Average} \times \text{New Number of Observations}) - (\text{Old Average} \times \text{Old Number of Observations})}{\text{New Number of Observations}}$ 

12 New Average = $\frac{\text{New Total Sum}}{\text{New Number of Observations}}$ 

PROFIT & LOSS

1. BASIC TERMS

CP = Cost Price
SP = Selling Price
MP = Marked Price

$$\text{Profit} = \text{SP} - \text{CP} \rightarrow \text{Profit}$$

$$\text{Loss} = \text{CP} - \text{SP} \rightarrow \text{Loss}$$

2. PROFIT & LOSS PERCENTAGE

$$\text{Profit \%} = \frac{\text{Profit}}{\text{CP}} \times 100$$

$$\text{Loss \%} = \frac{\text{Loss}}{\text{CP}} \times 100$$

$$\text{SP} = \text{CP} \times \left(1 + \frac{P}{100}\right) \rightarrow \text{Profit}$$

$$\text{SP} = \text{CP} \times \left(1 - \frac{L}{100}\right) \rightarrow \text{Loss}$$

3. WHEN SP & PROFIT % GIVEN

(Q) Profit = 20%, SP = 600
 $\rightarrow \text{CP} = 600 \times \frac{100}{120} = \underline{500}$ (Ans)

4. WHEN CP & LOSS % GIVEN

(Q) Loss = 25%, CP = 800
 $\rightarrow \text{SP} = 800 \times \frac{75}{100} = \underline{600}$ (Ans)

5. MARKED PRICE & DISCOUNT

$$\text{Discount} = \text{MP} - \text{SP}$$

$$\text{Discount \%} = \frac{\text{Discount}}{\text{MP}} \times 100$$

$$\text{SP} = \text{MP} \times \left(1 - \frac{d}{100}\right)$$

6. SUCCESSIVE DISCOUNTS

$$\text{Net discount \%} = a + b - \frac{ab}{100}$$

(Q) Discounts 20% & 10%
 $\text{Net discount} = 20 + 10 - \frac{20 \times 10}{100}$
 $= 30 - 2 = \underline{28\%}$ (Ans)

7. PROFIT WITH DISCOUNT (IMPORTANT)

(Q) MP = 1000, Discount = 20%, Profit = 25%
 Find CP.

$$\rightarrow \text{SP} = 1000 \times 0.8 = 800$$

$$\rightarrow \text{CP} = 800 \times \frac{100}{125} = \underline{640}$$
 (Ans)

8. FALSE WEIGHT PROBLEM PIT

If a trader uses less weight:

$$\text{Profit \%} = \frac{\text{Weight difference}}{\text{Actual weight}} \times 100$$

(Q) Uses 900g instead of 1kg

$$\text{Profit} = \frac{100}{900} \times 100 = \frac{100}{9} = \underline{11\frac{1}{9}\%}$$
 (Ans)

9. GAIN & LOSS ON SAME ITEM

(Q) Gain 10% on one item, Loss 10% on another.
 Net result?

→ Loss always

$$\text{Net loss \%} = \frac{a^2}{(100+a)^2} \times 100$$

For 10% → $\underline{0.83\%}$ loss

10. EQUAL SELLING PRICE CASE

(Q) Two articles sold at ₹100 each, Gain 20% on one, loss 20% on other.

$$\text{Net loss \%} = \frac{a^2}{100}$$

$$= \frac{20^2}{100} = \frac{400}{100} = \underline{4\% \text{ loss}}$$
 (Ans)

11. BUY x GET y FREE

$$\text{Effective discount \%} = \frac{y}{x+y} \times 100$$

(Q) Buy 2 get 1 free

$$\rightarrow \text{Discount} = \frac{1}{2+1} \times 100 = \underline{33\frac{1}{3}\%}$$
 (Ans)

12. SELLING ABOVE & BELOW CP

(Q) Selling at 20% profit instead of 10%,
 Extra profit = ₹100
 Find CP.

$$\text{Difference} = 20\% - 10\% = 10\%$$

$$\rightarrow \text{CP} = 100 \times \frac{100}{10} = \underline{1000}$$
 (Ans)

13. PARTNERSHIP LINK (IMPORTANT)

$$\text{Profit} \propto \text{Investment} \times \text{Time}$$

$$\text{Ratio of profits} = I_1 T_1 : I_2 T_2$$

PERCENTAGE CHEATSHEET

$$\frac{1}{3} = 33.33\%$$

$$\frac{2}{3} = 66.66\%$$

$$\frac{1}{4} = 25\%$$

$$\frac{2}{4} = 50\%$$

$$\frac{3}{4} = 75\%$$

$$\frac{1}{5} = 20\%$$

$$\frac{2}{5} = 40\%$$

$$\frac{3}{5} = 60\%$$

$$\frac{4}{5} = 80\%$$

$$\frac{1}{6} = 16.66\%$$

$$\frac{2}{6} = 33.33\%$$

$$\frac{3}{6} = 50\%$$

$$\frac{4}{6} = 66.66\%$$

$$\frac{5}{6} = 83.33\%$$

$$\frac{1}{7} = 14.28\%$$

$$\frac{2}{7} = 28.56\%$$

$$\frac{3}{7} = 42.84\%$$

$$\frac{4}{7} = 57.14\%$$

$$\frac{5}{7} = 71.42\%$$

$$\frac{6}{7} = 85.71\%$$

$$\frac{1}{8} = 12.5\%$$

$$\frac{2}{8} = 25\%$$

$$\frac{3}{8} = 37.50\%$$

$$\frac{4}{8} = 50\%$$

$$\frac{5}{8} = 62.50\%$$

$$\frac{6}{8} = 75\%$$

$$\frac{7}{8} = 87.50\%$$

$$\frac{1}{9} = 11.11\%$$

$$\frac{2}{9} = 22.22\%$$

$$\frac{3}{9} = 33.33\%$$

$$\frac{4}{9} = 44.44\%$$

$$\frac{5}{9} = 55.55\%$$

$$\frac{6}{9} = 66.66\%$$

$$\frac{7}{9} = 77.77\%$$

$$\frac{8}{9} = 88.88\%$$

$$\frac{1}{11} = 9.09\%$$

$$\frac{2}{11} = 18.18\%$$

$$\frac{3}{11} = 27.27\%$$

$$\frac{4}{11} = 36.36\%$$

$$\frac{5}{11} = 45.45\%$$

$$\frac{6}{11} = 54.54\%$$

$$\frac{7}{11} = 63.63\%$$

$$\frac{8}{11} = 72.72\%$$

$$\frac{9}{11} = 81.81\%$$

$$\frac{10}{11} = 90.90\%$$

$$\frac{1}{12} = 8.33\%$$

$$\frac{1}{13} = 7.69\%$$

$$\frac{1}{14} = 7.14\%$$

$$\frac{1}{15} = 6.66\%$$

$$\frac{1}{16} = 6.25\%$$

$$\frac{1}{24} = 4.16\%$$

$$\frac{1}{40} = 2.50\%$$

RATIO & PROPORTION

1. BASIC DEFINITIONS

Ratio = Comparison of two quantities of same kind

$$a : b = \frac{a}{b}$$

Proportion :

$$a : b :: c : d \Rightarrow \frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$$

2. PROPERTIES OF RATIO

- $a : b = ka : kb$
- $a : b = \frac{a}{k} : \frac{b}{k}$
- $a : b = b : a$ (inverse)

3. CONTINUED RATIO

$$\begin{aligned} a : b &= m : n \\ b : c &= p : q \\ \Rightarrow a : b : c &= mp : np : nq \end{aligned}$$

4. COMPOUND RATIO

$$(a : b) \times (c : d) = ac : bd$$

5. DIVIDING A QUANTITY IN A GIVEN RATIO

Sum = S, ratio = $a : b$

$$\text{First part} = S \times \frac{a}{a+b}$$

$$\text{Second part} = S \times \frac{b}{a+b}$$

(Q) Divide 300 in ratio 2 : 3

$$\Rightarrow \text{Total parts} = 2+3 = 5$$

$$1 \text{ part} = \frac{300}{5} = 60$$

$$\text{Parts} = 2 \times 60 = 120 \text{ \& } 3 \times 60 = 180$$

6. WHEN DIFFERENCE IS GIVEN

(Q) $a : b = 3 : 5$, difference = 20

$$\text{Let } a = 3x, b = 5x$$

$$5x - 3x = 20$$

$$2x = 20$$

$$x = 10$$

$$a = 3 \times 10 = 30, b = 5 \times 10 = 50$$

7. MULTIPLE RATIOS

(Q) $a : b = 2 : 3$

$$b : c = 4 : 5 ?$$

$$\text{LCM of } b (3 \& 4) = 12$$

$$a : b = (2 \times 4) : (3 \times 4) = 8 : 12$$

$$b : c = (4 \times 3) : (5 \times 3) = 12 : 15$$

$$\Rightarrow a : b : c = 8 : 12 : 15$$

8. WHEN SUM IS GIVEN (3 TERMS)

(Q) $a : b : c = 3 : 4 : 5$

$$\text{Sum} = 240$$

$$\text{Total parts} = 3+4+5 = 12$$

$$1 \text{ part} = \frac{240}{12} = 20 \Rightarrow a = 3 \times 20 = 60$$

$$b = 4 \times 20 = 80$$

$$c = 5 \times 20 = 100$$

9. COMPARISON TYPE

(Q) If $a : b = 3 : 2$

$$\text{Find } (4a + 5b) : (5a - 2b)$$

$$\text{Substitute } a = 3, b = 2$$

$$\Rightarrow (4 \times 3 + 5 \times 2) : (5 \times 3 - 2 \times 2)$$

$$\Rightarrow (12 + 10) : (15 - 4)$$

$$\Rightarrow 22 : 11 = 2 : 1$$

10. PROPORTIONAL DIVISION

$$\text{If } x \propto y \Rightarrow \frac{x}{y} = \text{constant}$$

(Q) If $x : y = 5 : 7$ and $x + y = 144$

$$\text{Total parts} = 5+7 = 12$$

$$1 \text{ part} = \frac{144}{12} = 12 \Rightarrow x = 5 \times 12 = 60$$

$$y = 7 \times 12 = 84$$

11. IMPORTANT IDENTITIES

If $a : b = m : n$

$$\bullet (a+b) : (a-b) = (m+n) : (m-n)$$

$$\bullet (a^2 - b^2) : (ab) = (m^2 - n^2) : (mn)$$

12. MIXED POWER RATIOS

(Q) If $a : b = 2 : 3$

$$\text{Find } a^2 : b^2 = 2^2 : 3^2 = 4 : 9$$

$$\text{Find } \sqrt{a} : \sqrt{b} = \sqrt{2} : \sqrt{3}$$

13. VARIATION

$$\text{Direct Variation } x \propto y \Rightarrow x = ky$$

$$\text{Inverse Variation } x \propto \frac{1}{y} \Rightarrow xy = k$$

14. IMPORTANT SHORTCUTS

- ✓ Always convert to same units
- ✓ Use substitution for complex expressions
- ✓ If powers differ $\rightarrow$ cannot compare directly
- ✓ LCM trick saves time in chain ratios

15. COMMON MISTAKES

- ✗ Ignoring units
- ✗ Wrong LCM in chain ratio
- ✗ Mixing sum & difference logic
- ✗ Wrong unit conversion in divisibility

SQUARE NUMBERS

[1 to 100]

$1^2 = 1$
 $2^2 = 4$
 $3^2 = 9$
 $4^2 = 16$
 $5^2 = 25$
 $6^2 = 36$
 $7^2 = 49$
 $8^2 = 64$
 $9^2 = 81$
 $10^2 = 100$
 $11^2 = 121$
 $12^2 = 144$
 $13^2 = 169$
 $14^2 = 196$
 $15^2 = 225$
 $16^2 = 256$
 $17^2 = 289$
 $18^2 = 324$
 $19^2 = 361$
 $20^2 = 400$
 $21^2 = 441$
 $22^2 = 484$
 $23^2 = 529$
 $24^2 = 576$
 $25^2 = 625$

$26^2 = 676$
 $27^2 = 729$
 $28^2 = 784$
 $29^2 = 841$
 $30^2 = 900$
 $31^2 = 961$
 $32^2 = 1024$
 $33^2 = 1089$
 $34^2 = 1156$
 $35^2 = 1225$
 $36^2 = 1296$
 $37^2 = 1369$
 $38^2 = 1444$
 $39^2 = 1521$
 $40^2 = 1600$
 $41^2 = 1681$
 $42^2 = 1764$
 $43^2 = 1849$
 $44^2 = 1936$
 $45^2 = 2025$
 $46^2 = 2116$
 $47^2 = 2209$
 $48^2 = 2304$
 $49^2 = 2401$
 $50^2 = 2500$

$51^2 = 2601$
 $52^2 = 2704$
 $53^2 = 2809$
 $54^2 = 2916$
 $55^2 = 3025$
 $56^2 = 3136$
 $57^2 = 3249$
 $58^2 = 3364$
 $59^2 = 3481$
 $60^2 = 3600$
 $61^2 = 3721$
 $62^2 = 3844$
 $63^2 = 3969$
 $64^2 = 4096$
 $65^2 = 4225$
 $66^2 = 4356$
 $67^2 = 4489$
 $68^2 = 4624$
 $69^2 = 4761$
 $70^2 = 4900$
 $71^2 = 5041$
 $72^2 = 5184$
 $73^2 = 5329$
 $74^2 = 5476$
 $75^2 = 5625$

$76^2 = 5776$
 $77^2 = 5929$
 $78^2 = 6084$
 $79^2 = 6241$
 $80^2 = 6400$
 $81^2 = 6561$
 $82^2 = 6724$
 $83^2 = 6889$
 $84^2 = 7056$
 $85^2 = 7225$
 $86^2 = 7396$
 $87^2 = 7569$
 $88^2 = 7744$
 $89^2 = 7921$
 $90^2 = 8100$
 $91^2 = 8281$
 $92^2 = 8464$
 $93^2 = 8649$
 $94^2 = 8836$
 $95^2 = 9025$
 $96^2 = 9216$
 $97^2 = 9409$
 $98^2 = 9604$
 $99^2 = 9801$
 $100^2 = 10000$

PROBABILITY



1. BASIC DEFINITION

Probability of an event E :

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

2. SAMPLE SPACE

Sample space (S) → Set of all possible outcomes

(Q) Tossing a coin

$$\rightarrow S = \{H, T\}$$

(Q) Rolling a die

$$\rightarrow S = \{1, 2, 3, 4, 5, 6\}$$

3. RANGE OF PROBABILITY

$$0 \leq P(E) \leq 1$$

• $P(E) = 0 \rightarrow$ Impossible event

• $P(E) = 1 \rightarrow$ Certain event

4. COMPLEMENTARY EVENTS

If E is an event,

$$P(E') = 1 - P(E)$$

(Q) Probability of NOT getting a head

$$\rightarrow = 1 - P(H)$$

5. ADDITION THEOREM

For any two events A and B :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Mutually Exclusive Events

If $A \cap B = \emptyset$,

$$P(A \cup B) = P(A) + P(B)$$

6. INDEPENDENT EVENTS

Two events A and B are independent if :

$$P(A \cap B) = P(A) \times P(B)$$

(Q) Tossing two coins

$$\rightarrow P(HH) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

7. CONDITIONAL PROBABILITY

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

(Q) Probability of A given B has occurred.

8. BAYES' THEOREM

If $A_1, A_2, \dots, A_n$ are mutually exclusive events and B is an event :

$$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{\sum P(A_j)P(B|A_j)}$$

⇒ Always draw tree diagram



9. PROBABILITY IN CARDS



• Face cards = 12, Kings = 4, Queens = 4, Aces = 4

(Q) Probability of a king $\rightarrow \frac{4}{52} = \frac{1}{13}$

10. PROBABILITY IN DICE

• 1 die $\rightarrow$ 6 outcomes

• 2 dice $\rightarrow$ 36 outcomes

(Q) Sum = 7

$$\rightarrow \text{Favourable} = 6; P = \frac{6}{36} = \frac{1}{6}$$

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

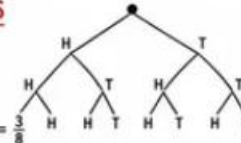
11. PROBABILITY IN COINS

• 1 coin $\rightarrow$ 2 outcomes

• 2 coins $\rightarrow$ 4 outcomes

• 3 coins $\rightarrow$ 8 outcomes

(Q) Exactly 2 heads in 3 tosses = $\frac{3}{8}$



12. AT LEAST / AT MOST CONCEPT

$$\text{"At least one"} = 1 - (\text{None})$$

(Q) At least one head in 2 tosses

$$\rightarrow = 1 - P(TT) = 1 - \frac{1}{4} = \frac{3}{4}$$

13. IMPORTANT SHORTCUTS

- ✓ Use complement for "at least"
- ✓ Independent $\rightarrow$ multiply
- ✓ Mutually exclusive $\rightarrow$ add
- ✓ Always write sample space

14. COMMON MISTAKES

- ✗ Forgetting total outcomes
- ✗ Confusing independent & exclusive
- ✗ Not subtracting intersection

SIMPLE INTEREST & COMPOUND INTEREST

1. BASIC TERMS

P = Principal
R = Rate of Interest (% per annum)
T = Time (years)
SI = Simple Interest
CI = Compound Interest
A = Amount

2. SIMPLE INTEREST (SI)

Formula

$$SI = \frac{P \times R \times T}{100}$$

$$\text{Amount} = P + SI$$

Example

P = ₹1000, R = 10%, T = 2 years

$$SI = \frac{1000 \times 10 \times 2}{100} = \underline{200}$$

$$\text{Amount} = 1000 + 200 = \underline{1200}$$

3. SHORTCUT METHOD (SI)

Interest for 1 year = R% of P

Interest for T years = T × (R% of P)

4. WHEN AMOUNT GIVEN AT DIFFERENT TIMES (SI)

Amount after 3 yrs = ₹900

Amount after 6 yrs = ₹1200

Find P & R.

- Extra interest for 3 yrs = 1200 - 900 = 300
- 1 year interest = $\frac{300}{3} = \underline{100}$
- P = Amount (3 yrs) - Interest (3 yrs)
= 900 - 300 = 600
- R = $\frac{\text{Interest for 1 yr}}{P} \times 100 = \frac{100}{600} \times 100 = \frac{100}{6} = \underline{16\frac{2}{3}\%}$

5. TIME FOR MONEY TO DOUBLE (SI)

$$T = \frac{100}{R}$$

6. COMPOUND INTEREST (CI)

Formula (Yearly)

$$A = P \left(1 + \frac{R}{100}\right)^T$$

$$CI = A - P$$

Example

P = 1000, R = 10%, T = 2

$$A = 1000 \left(1 + \frac{10}{100}\right)^2 = 1000 (1.1)^2$$

$$= 1000 \times 1.21 = \underline{1210}$$

$$CI = 1210 - 1000 = \underline{210}$$

7. HALF-YEARLY / QUARTERLY COMPOUNDING

Half-Yearly

$$\text{Rate} = \frac{R}{2}$$

$$\text{Time} = 2T$$

Quarterly

$$\text{Rate} = \frac{R}{4}$$

$$\text{Time} = 4T$$

$$A = P \left(1 + \frac{R}{200}\right)^{2T}$$

$$A = P \left(1 + \frac{R}{400}\right)^{4T}$$

8. DIFFERENCE BETWEEN CI & SI (IMPORTANT)

$$\text{For 2 years: } CI - SI = P \left(\frac{R}{100}\right)^2$$

(Q) Difference = 25, R = 10%

$$P = \frac{\text{Difference} \times 100^2}{R^2} = \frac{25 \times 100 \times 100}{10 \times 10} = \underline{2500}$$

9. WHEN MONEY BECOMES MULTIPLE TIMES (CI)

If money doubles:

$$2P = P \left(1 + \frac{R}{100}\right)^T$$

If money becomes 8 times:

$$8 = 2^3 \Rightarrow T = 3 \times (\text{doubling time})$$

10. COMPARISON: SI vs CI

- CI > SI (always, except T = 1 year)
- Difference increases with time
- CI grows exponentially

11. IMPORTANT SHORTCUTS

- ✓ For SI → Linear growth
- ✓ For CI → Exponential growth
- ✓ Use difference formula for 2-year questions
- ✓ Convert months → years carefully



= NUMBERS =

- ⦿ **Natural numbers** : 1, 2, 3, 4, 5, ...
- ⦿ **Even numbers** : 2, 4, 6, 8, 10, ...
- ⦿ **Odd numbers** : 1, 3, 5, 7, 9, ...
- ⦿ **Integer numbers** : -3, -2, -1, 0, 1, 2, 3, ...
- ⦿ **Whole numbers** : 0, 1, 2, 3, 4, ...
- ⦿ **Composite numbers** : 4, 6, 8, 9, ...
- ⦿ **Prime numbers** : 2, 3, 5, 7, 11, ...
- ⦿ **Co-prime numbers** : (5, 7), (2, 3)
- ⦿ **Rational numbers** : $\sqrt{4}$, $\frac{7}{5}$, $\frac{2}{3}$, 3
- ⦿ **Irrational numbers** : $\sqrt{5}$, $\sqrt{7}$, $\sqrt{11}$, $\sqrt{13}$, π
- ⦿ **Real numbers** : $\sqrt{4}$, $11\frac{4}{7}$
- ⦿ **Surreal numbers** : $\sqrt{-6}$, $\sqrt{-5}$, $\sqrt{-29}$
- ⦿ The smallest prime number is : **2 (two)**
- ⦿ The smallest even prime number is : **2 (two)**
- ⦿ The smallest composite number is : **4 (four)**
- ⦿ The smallest whole number is : **0 (zero)**
- ⦿ The smallest natural number is : **1 (one)**
- ⦿ π (pi) : **Irrational number**

⦿ π (pi) is approximately : $\frac{22}{7}$ or 3.14159

NUMBER SYSTEM

1. TYPES OF NUMBERS

- Natural Numbers (N) : 1, 2, 3, ...
- Whole Numbers (W) : 0, 1, 2, 3, ...
- Integers (Z) : ... -3, -2, -1, 0, 1, 2, 3 ...
- Rational Numbers (Q) : $\frac{p}{q}$, $q \neq 0$
- Irrational Numbers : $\sqrt{2}$, $\sqrt{3}$, π
- Real Numbers (R) : Rational + Irrational

2. EVEN & ODD NUMBERS

$$\begin{aligned}\text{Even} &= 2n \\ \text{Odd} &= 2n + 1\end{aligned}$$

- Even + Even = Even
- Odd + Odd = Even
- Even + Odd = Odd
- Odd × Odd = Odd
- Even × Any = Even

3. PRIME & COMPOSITE NUMBERS

- Prime → Exactly 2 factors (1 & itself)
- Composite → More than 2 factors
- Smallest prime = 2
- Only even prime = 2

4. FACTORS & MULTIPLES

- Factor → Divides exactly
- Multiple → Product of number

(Q) Number of factors of $n =$

$$\text{If } n = p^a q^b r^c$$

$$\text{Total factors} = [(a+1)(b+1)(c+1)]$$

5. HCF & LCM

- HCF → Lowest power of common primes
- LCM → Highest power of all primes

$$\text{HCF} \times \text{LCM} = \text{Product of two numbers}$$

(Q) If HCF = 12, LCM = 360

$$\begin{aligned}\text{Numbers} &= 12x, 12y \\ 12x \times 12y &= 12 \times 360 \\ \rightarrow xy &= 30\end{aligned}$$

6. DIVISIBILITY RULES

- 2 → Last digit even
- 3 → Sum of digits divisible by 3
- 4 → Last two digits divisible by 4
- 5 → Last digit 0 or 5
- 6 → Divisible by 2 & 3
- 8 → Last three digits divisible by 8
- 9 → Sum of digits divisible by 9
- 11 → (Sum of odd place - even place) divisible by 11

7. REMAINDER THEOREM

If n divided by d gives remainder r

$$n = dq + r$$

(Q) Find remainder when 7^{25} divided by 6

$$\begin{aligned}7 &\equiv 1 \pmod{6} \\ \Rightarrow 7^{25} &\equiv 1^{25} \pmod{6} \\ \Rightarrow \text{Remainder} &= 1 \text{ (Ans)}\end{aligned}$$

8. CYCLICITY (LAST DIGIT)

Last digits repeat in cycles of 4

- 2 → 2, 4, 8, 6
- 3 → 3, 9, 7, 1
- 7 → 7, 9, 3, 1

(Q) Find last digit of 7^{101}

$$\begin{aligned}101 \text{ mod } 4 &= 1 \\ \Rightarrow \text{Last digit} &= 7^1 = 7 \text{ (Ans)}\end{aligned}$$

9. FINDING NUMBER USING HCF & LCM

(Q) Two numbers have HCF = 6
If one number = 30
Find other number
Other = $\frac{6 \times 180}{30} = 6 \times 6 = 36$.

11. DECIMAL TO FRACTION

$$\begin{aligned}\text{Let } x &= 0.333... \\ 10x &= 3.333... \\ 9x &= 3 \\ x &= \frac{3}{9} = \frac{1}{3} \text{ (Ans)}\end{aligned}$$

13. SIMPLIFICATION TRICKS

- ✓ Use prime factorization
- ✓ Cyclicity saves time
- ✓ Last digit saves time

15. COMMON MISTAKES

- ✗ Ignoring sign of integers
- ✗ Wrong cyclic power
- ✗ Mixing HCF & LCM rules
- ✗ Wrong conversion in divisibility

10. FRACTIONS

- Proper : numerator < denominator
- Improper : numerator ≥ denominator
- Convert mixed to improper

$$a \frac{b}{c} = \frac{ac + b}{c}$$

12. SURDS

$$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$$

$$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}} = 5\sqrt{2}$$

13. SIMPLIFICATION TRICKS

- BODMAS rule
- Always simplify to fractions if needed

IMPORTANT SHORTCUTS

- ✓ Use prime factorization
- ✓ HCF × LCM trick
- ✓ Last digit & remainder checks